Operations with Standard Form

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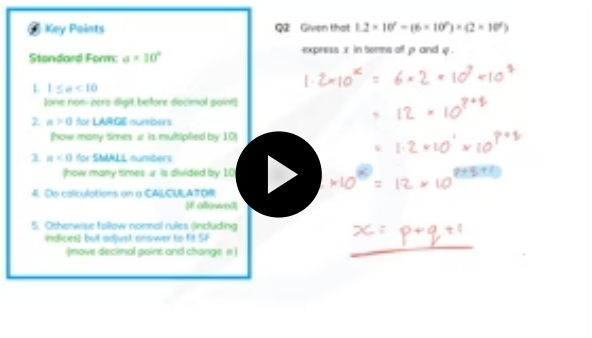

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Operations with standard form

How do I perform calculations in standard form using a calculator?

- Make **use of brackets** around each number, and use the $\boxed{\times 10^x}$ button to enter numbers in standard form
  - e.g. $(3 \times 10^8) \times (2 \times 10^{-3})$
  - You can instead use the standard multiplication and index buttons
- If your calculator answer is **not in standard form**, but the question requires it:
  - Either **rewrite** it using the standard process
    - e.g. $3\,820\,000 = 3.82 \times 10^6$
  - Or rewrite numbers in standard form, then apply the **laws of indices**
    - e.g. $243 \times 10^{20} = (2.43 \times 10^2) \times 10^{20} = 2.43 \times 10^{22}$

How do I multiply and divide numbers in standard form without a calculator?

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **STEP 1**
  Multiply the **ordinary** numbers
  - e.g. $3 \times 4 = 12$

- **STEP 2**
  Multiply the **powers** of 10 by **adding** the indices
  - e.g. $10^2 \times 10^5 = 10^{2+5} = 10^7$

- **STEP 3**
  Multiply the new ordinary number and the new power of 10
  - e.g. 12×10^7

- **STEP 4 (if needed)**
  Write the ordinary number in **standard form** and simplify the powers of 10 by **adding** the indices
  - e.g. $12 = 1.2 \times 10^1$ and $1.2 \times 10^1 \times 10^7 = 1.2 \times 10^8$

Division

- Consider the example $(2 \times 10^{-5}) \div (8 \times 10^{-3})$
- **STEP 1**
  Divide the **first ordinary** number by the **second ordinary** number
  - e.g. $2 \div 8 = 0.25$

- **STEP 2**
  Divide the **first power** of 10 by the **second power** of 10 by **subtracting** the indices
  - e.g. $10^{-5} \div 10^{-3} = 10^{-5-(-3)} = 10^{-2}$
    - Be careful with negatives!

- **STEP 3**
  Multiply the new ordinary number and the new power of 10
  - e.g. 0.25×10^{-2}

- **STEP 4 (if needed)**
  Write the ordinary number in **standard form** and simplify the powers of 10 by **adding** the indices
  - e.g. $0.25 = 2.5 \times 10^{-1}$ and
    $2.5 \times 10^{-1} \times 10^{-2} = 2.5 \times 10^{-3}$

How do I add and subtract numbers in standard form without a calculator?

Method 1

- Consider the example $(3.2 \times 10^3) + (2.1 \times 10^2)$
- **STEP 1**
  Convert both numbers to **ordinary** numbers
  - e.g. $3.2 \times 10^3 = 3200$ and $2.1 \times 10^2 = 210$

- **STEP 2**
  Add or **subtract** the ordinary numbers
  - e.g. $3200 + 210 = 3410$

- **STEP 3 (if needed)**
  Convert the answer to **standard form**
  - e.g. $3410 = 3.41 \times 10^3$

Method 2

has the same power of 10 as the number with the **lowest power** of 10
  - e.g. $4 \times 10^{50} = 4 \times 10^{2} \times 10^{48} = 400 \times 10^{48}$
    - The ordinary gets bigger as the power of 10 gets smaller

- **STEP 2**
  Collect like terms by adding or subtracting the **ordinary** numbers
  - e.g. $(400 \times 10^{48}) - (2 \times 10^{48}) = 398 \times 10^{48}$
    - Do not change the power of 10

- **STEP 3**
  Write the ordinary number in **standard form** and simplify the powers of 10 by **adding** the indices
  - e.g. $398 = 3.98 \times 10^{2}$ and
    $3.98 \times 10^{2} \times 10^{48} = 3.98 \times 10^{50}$

- This method works for **negative powers** too
  - e.g. consider $(8 \times 10^{-20}) - (5 \times 10^{-21})$
    - $8 \times 10^{-20} = 8 \times 10^{1} \times 10^{-21} = 80 \times 10^{-21}$
    - $(80 \times 10^{-21}) - (5 \times 10^{-21}) = 75 \times 10^{-21}$
    - $75 = 7.5 \times 10^{1}$ and $7.5 \times 10^{1} \times 10^{-21} = 10^{-20}$

Examiner Tips and Tricks

The second method is the most efficient when the powers of 10 have large positive or large negative indices.

Worked Example

Without using a calculator, find $(45 \times 10^{-3}) \div (0.9 \times 10^{5})$.

Write your answer in the form $A \times 10^{n}$, where $1 \leq A < 10$ and n is an integer.

Answer:

Rewrite the division as a fraction, then separate out the powers of 10

$$\frac{45 \times 10^{-3}}{0.9 \times 10^{5}} = \frac{45}{0.9} \times \frac{10^{-3}}{10^{5}}$$

Work out $\frac{45}{0.9}$

$$\frac{45}{0.9} = \frac{450}{9} = 50$$

Work out $\frac{10^{-3}}{10^{5}}$ using laws of indices

$$\frac{10^{-3}}{10^{5}} = 10^{-3-5} = 10^{-8}$$

Rewrite in standard form, where a is between 1 and 10

$$50 \times 10^{-8} = 5 \times 10 \times 10^{-8} = 5 \times 10^{-7}$$

$$5 \times 10^{-7}$$

Worked Example

Without using a calculator, find $(2.8 \times 10^{-6}) + (9.7 \times 10^{-8})$.

Write your answer in the form $A \times 10^{n}$, where $1 \leq A < 10$ and n is an integer.

Answer:

Rewrite 2.8×10^{-6} so that the power of 10 matches 9.7×10^{-8}

$$2.8 \times 10^{-6} = 280 \times 10^{-8}$$

The numbers can now be added together, keeping the power of 10 the same

$$(280 \times 10^{-8}) + (9.7 \times 10^{-8}) = 289.7 \times 10^{-8}$$

Write the number in standard form

$$2.897 \times 10^{2} \times 10^{-8} = 2.897 \times 10^{2-8}$$

$$2.897 \times 10^{-6}$$

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